Moving Big Data: Pipeline Orchestration

Building a data pipeline is only half the challenge — keeping it running reliably, on time, and in the right order is where orchestration comes in. In this lesson, we explore how data engineers use triggers, schedules, and event-driven logic to automate complex workflows at scale. From cron-based scheduling to serverless event-driven architectures, you will learn how to design pipelines that run themselves while remaining observable and maintainable.



What You'll Learn:

Schedule-based Orchestration

Understand how time-based triggers and cron expressions are used to automate recurring pipeline runs, and how tools like Apache Airflow manage dependencies between tasks.

Event-based Orchestration

Explore how pipelines can be triggered by events — such as a file landing in object storage or a message arriving on a queue — rather than fixed schedules, enabling more responsive and efficient data flows.

Serverless and Small-scale Processing

Learn how serverless functions (e.g. AWS Lambda, GCP Cloud Functions) can handle lightweight transformations and integration tasks without managing infrastructure, and when this approach is preferable to a full orchestration platform.

Triggers, Schedules, and Pipeline Design Trade-offs

Compare schedule-driven and event-driven approaches across real-world scenarios, and understand how to choose the right orchestration strategy based on latency requirements, data volume, and system complexity.

Ready to Dive In?

Reflect:
Think of a repetitive task in your own life that runs on a schedule — a weekly report, a monthly payment, a daily backup. Now imagine it needed to react to a specific event instead. How would the logic change, and what could go wrong if the timing or order wasn't handled correctly?